Symplectic topology

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Lecture 1

1 Motivation

Symplectic topology is the study of global phenomenon of *symplectic manifolds*, which are a pair (M, ω), where M is a smooth manifold, and ω is a non-degenerate closed 2-form on M. In particular, non-generacy implies the map

$$\begin{aligned} TM &\to T^*M \\ v &\mapsto \omega(v, \cdot) \end{aligned}$$

is an isomorphism.

Example 1.1

$\mathbb{R}^{2n}$ with coordinates $x_1, y_1, \ldots, x_n, y_n$ is a symplectic manifold, with

$$\omega_{\text{std}} = \omega_0 = \sum \mathrm{d}x_i \wedge \mathrm{d}y_i$$

Example 1.2

Let Σ be a surface of genus g. Then any volume form makes Σ_g into a symplectic manifold.

Why study such manifolds? Some motivation from classical mechanics: Say we have a phase space X encoding position and momentum of some particles in a physical system. For example, we have a pendulum with angular coordinate θ, and angular momentum $p = m\dot{\theta}$. The phase space is then $S^1 \times \mathbb{R}$.

Given $H : X \to \mathbb{R}$, encoding the total energy of a configuration, we want a vector field $\mathcal{X}_H$ such that the trajectory $x(t)$ satisfies

$$\dot{x}(t) = \mathcal{X}_H(x(t))$$

It's not enough to know that H is constant on trajectories, i.e. $\mathrm{d}H(\mathcal{X}_H) = 0$, since this is a co-dimension 1 condition. Recall if X has symplectic form ω, then we can take $\mathcal{X}_H$ such that

$$\omega(\mathcal{X}_H, \cdot) = \mathrm{d}H \tag{1}$$

Indeed, every phase space has such a symplectic form. Note that ω is non-degenerate if and only if eq. (1) has a solution for all H, and ω being closed is that ω is preserved under the flow of $\mathcal{X}_H$.

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Example 1.3 (harmonic oscillator)
In this case, $X = \mathbb{R}^2$, with coordinates p for momentum, q for position. The symplectic form is

$$dp \wedge dq = r dr \wedge d\theta$$

where (r, θ) is polar coordinates on X. This has Hamiltonian

$$H = \frac{1}{2}(p^2 + q^2) = \frac{1}{2}r^2$$

Hence

$$dH = r dr$$

Thus, $\mathcal{X}_H = \frac{\partial}{\partial \theta}$.

1.1 Mathematical perspective

A question to keep in mind is: How does symplectic geometry compare to e.g. Riemannian geometry, complex geometry etc.

In this context, $d\omega = 0$ is a form of "flatness". Suppose (Y, g) is a Riemannian manifold, i.e. we have $g \in \mathrm{Sym}^2 T^*Y$ non-degenerate. It is a fact that (Y, g) is locally isometric to $\mathbb{R}^n$ with the Euclidean metric if and only if the curvature F of g vanishes. That is, g is flat.

On the other hand, suppose (Z, J) is an almost complex manifold. That is, $J \in \mathrm{End}(TZ)$, with $J^2 = -\mathrm{id}_{TZ}$. One example would be $\mathbb{C}^n$, with $J(z) = iz$. One question is, when is (Z, J) locally isomorphic to $\mathbb{C}^n$? That is, when is (Z, J) a complex manifold. A difficult result in analysis, called the Newlander-Nirenberg theorem, shows that this occurs if and only if the Nijenhuis tensor,

$$\mathcal{N}_J(v, w) = [Jv, Jw] - J([v, Jw] + [Jv, w]) - [v, w]$$

vanishes

Theorem 1.4 (Darboux). A manifold (X, ω) with non-degenerate 2-form ω, is locally modeled on $(\mathbb{R}^{2n}, \omega_{std})$ if and only if $d\omega = 0$.

Finally, symplectic topology is ubiquitous in research in geometry and topology, e.g. global topology, algebraic geometry and so on.

Lecture 2

2 (Symplectic) linear algebra

Let V be a vector space, over a field $\mathbb{F}$ of charactistic 0, with $\dim(V) = n$. $V^* = \mathrm{Hom}(V, \mathbb{F})$ will denote the dual.

Define $\mathbb{F}$-algebras

$$T(V) = \bigoplus_{i=0}^{\infty} V^{\otimes i}$$
$$\Lambda^*(V) = \frac{T(V)}{\langle v \otimes v \rangle}$$

If $\{e_1, \ldots, e_n\}$ is a basis for V, then

$$\{e_{i_1} \wedge \cdots \wedge e_{i_k} \mid i_1 < \cdots < i_k\}$$

is a basis for $\Lambda^k V$. Using this,

$$\dim(\Lambda^k V) = \binom{n}{k}$$

Proposition 2.1. $\Lambda^k(V^*)$ is isomorphic to the space of alternating multilinear maps

$$V^{\otimes k} \to \mathbb{F}$$

Proof. The map is given by

$$e_{i_1}^\vee \wedge \cdots \wedge e_{i_k}^\vee \mapsto \sum_{\sigma \in S_k} \operatorname{sign}(\sigma) e_{i_{\sigma(1)}}^\vee \otimes \cdots \otimes e_{i_{\sigma(k)}}^\vee$$

The rest is left as an exercise. □

In particular, $\Lambda^n(V^*) = \Lambda^{\text{top}}(V^*)$ is (non-canonically) isomorphic to $\operatorname{span}\{\det\}$, or volume forms.

Definition 2.2

A *symplectic form* on V is $\omega \in \Lambda^2(V^*)$, such that $\omega : V \times V \to \mathbb{F}$ is a perfect pairing[a].

[a] That is, $\omega(v, \cdot)$ induces an isomorphism $V \cong V^*$.

Example 2.3

Let M^{4m+2} be an oriented manifold, and $V = H^{2m+1}(M; \mathbb{Q})$. The pairing

$$\langle \alpha, \beta \rangle = \int_M \alpha \cup \beta$$

is a symplectic form.

Example 2.4

Let $V = \mathbb{R}^{2n}$, with basis $\{x_i, y_i\}$, with dual basis $\{e_i, f_i\}$. We can define

$$\omega = \sum_{i=1}^{n} e_i \wedge f_i$$

More generally, if $(V, \omega_V), (W, \omega_W)$ are symplectic, then so is

$$(V \oplus W, \omega_V \oplus \omega_W)$$

We can write

$$\omega(v, w) = v \cdot J_0 w$$

where

$$J_0 e_j = f_j \quad \text{and} \quad J_0 f_j = -e_j$$

If we use complex coordinates $z_j = x_j + i y_j$, then J_0 is just multiplication bu i.

Let (V, ω) be a symplectic vector space. Let $W \subseteq V$ be a subspace. Define

$$W^\omega = \{v \in V \mid \omega(w, v) = 0 \text{ for all } w \in W\}$$

Definition 2.5

W is

1. *isotropic* if $W \subseteq W^\omega$,
2. *coisotropic* if $W^\omega \subseteq W$,

3. *symplectic* if $W \cap W^\omega = 0$,
4. *Lagrangian* if $W = W^\omega$. That is, it is isotropic and coisotropic.

Example 2.6

Take $\mathbb{R}^{2n}$ with the standard symplectic form

$$\omega = \sum_i \mathrm{d}x_i \wedge \mathrm{d}y_i$$

Then $\mathrm{span}\{x_1, x_2\}$ is isotropic. In fact, $\mathrm{span}\{x_1, \ldots, x_n\}$ is a Lagrangian subspace. On the other hand, $\mathrm{span}\{x_1, y_1\}$ is a symplectic subspace.

Remark 2.7. For *any* subspace $W \subseteq V$,

$$\dim(W) + \dim(W^\omega) = \dim(V)$$

and we have that $(W^\omega)^\omega = W$.

Theorem 2.8. Given any symplectic vector space (V, ω) over $\mathbb{R}$, there exists a linear isomorphism $\Phi : V \to \mathbb{R}^{2n}$, such that $\Phi^*\omega_0 = \omega$.

Proof. We induct on $\dim(V)$. Choose any $v_1 \in V$ non-zero. By non-degeneracy, there exists $w_1 \in V$ such that $\omega(v_1, w_1) = 1$. Let $V_1 = \mathrm{span}\{v_1, w_1\}$. Define $\Phi_1 : V_1 \to \mathbb{R}^2$, by $\Phi_1(v_1) = x_1$ and $\Phi_1(w_1) = y_1$. This satisfies that $\Phi_1^*\omega_0 = \omega|_{V_1}$.

Define $V_2 = V_1^\omega$. Then V_1 and V_2 are symplectic, and $\dim(V_2) = \dim(V) - 2$. By induction hypothesis, there exists $\Phi_2 : V_2 \to \mathbb{R}^{2n-2}$, with $\Phi_2^*\omega_0 = \omega|_{V_2}$. We can then define

$$\Phi = \Phi_1 \oplus \Phi_2 : (V_1, \omega|_{V_1}) \oplus (V_2, \omega|_{V_2}) \to (\mathbb{R}^{2n-2}, \omega_0) \oplus (\mathbb{R}^2, \omega_0) = (\mathbb{R}^{2n}, \omega_0)$$

□

Lecture 3

In particular, symplectic vector spaces are always even dimensional.

Corollary 2.9. $\alpha \in \Lambda^2 V^*$ is symplectic if and only if $\alpha \wedge \cdots \wedge \alpha \in \Lambda^{\mathrm{top}} V^* = \mathbb{R}$ is non-zero.

Proof. Note that

$$\omega_{\mathrm{std}}^n = x_1^\vee \wedge y_1^\vee \cdots \wedge x_n^\vee \wedge y_n^\vee$$

is the usual volume form. □

Returning to linear algebra, we have

$$\mathrm{Hom}(V, V) \to \mathrm{Hom}(\Lambda^{\mathrm{top}} V, \Lambda^{\mathrm{top}} V) \cong \mathbb{R}$$

The composition is the *determinant* $\det : \mathrm{Hom}_*(V, V) \to \mathbb{R}$.

2.1 Orientations

Define

$$\mathrm{or}_V = \frac{\{\text{ordered bases for } V\}}{\mathrm{GL}_+(V)} = \frac{\mathrm{Fr}(V)}{\mathrm{GL}_+(V)}$$

where we consider the $\mathrm{GL}_+(V)$-orbits, i.e. two frames are equivalent if they are related by a transformation with positive determinant.

In this case, or_V has two elements. Given any volume form α on V, there exists an induced orientation on V, given by the sign of α on a basis. In particular, any symplectic vector space has such a canonical orientation, induced by ω^n.

2.2 Symplectomorphism

Definition 2.10

Define

$$\mathrm{Sp}(2n) = \{\Phi \in \mathrm{GL}(\mathbb{R}^{2n}) \mid \Phi^* \omega_{\mathrm{std}} = \omega_{\mathrm{std}}\}$$

if J_0 is the matrix so that $\omega(u, v) = u \cdot J_0 v$, then we have that

$$\mathrm{Sp}(2n) = \{\Phi \in \mathrm{GL}(\mathbb{R}^{2n}) \mid \Phi^{\mathsf{T}} J_0 \Phi = J_0\}$$

Example 2.11

For $\mathbb{R}^2$, ω is the standard volume form $\mathrm{d}x \wedge \mathrm{d}y$. So preserving ω is the same as preserving the volume form. Hence

$$\mathrm{Sp}(2) = \mathrm{SL}(2, \mathbb{R})$$

In general, we have an embedding

$$\mathrm{Sp}(2n) \subseteq \mathrm{SL}(2n, \mathbb{R})$$

since ω^n is the standard volume form[a] on $\mathbb{R}^{2n}$.

[a] maybe up to a factor of $n!$

Note if Φ commutes with J_0, and Φ preserves the inner product, then $\Phi \in \mathrm{Sp}(2n)$. So

$$\mathrm{GL}(n, \mathbb{C}) \cap \mathrm{O}(2n) \subseteq \mathrm{Sp}(2n)$$

In fact, we have

$$\mathrm{GL}(n, \mathbb{C}) \cap \mathrm{O}(2n) = \mathrm{GL}(n, \mathbb{C}) \cap \mathrm{Sp}(2n) = \mathrm{Sp}(2n) \cap \mathrm{O}(2n) = \mathrm{U}(n)$$

Proposition 2.12. The set S of symplectic forms on $\mathbb{R}^{2n}$, as a subset of $\Lambda^2(V^*)$ is $\mathrm{GL}(2n, \mathbb{R})/\mathrm{Sp}(2n)$, a homoegenous space.

Proof. $\mathrm{GL}(2n, \mathbb{R})$ acts transitively on S, with stabiliser $\mathrm{Sp}(2n)$ for any point. □

Let $\mathrm{Lag}(V)$ be the space of Lagrangian subspaces L of V.

Example 2.13

$\mathrm{Lag}(\mathbb{R}^2)$ is just the one-dimensional subspaces of $\mathbb{R}^2$, i.e. $\mathbb{RP}^1 = S^1$. More generally,

$$\mathrm{Lag}(\mathbb{R}^{2n}) = \frac{\mathrm{U}(n)}{\mathrm{O}(n)}$$

Example 2.14

Let $V = C^\infty(S^1, \mathbb{R})$, with

$$\omega(f, g) = \int_{S^1} f \mathrm{d}g$$

Using integration by parts, this is alternating.

2.3 Vector bundles

If $\pi : \mathcal{E} \to X$ is a vector bundle, we can define $\Lambda^* \mathcal{E} \to X$, $\Lambda^*(\mathcal{E}^*) \to X$, $\mathrm{or}_{\mathcal{E}} \to X$, and so on. We have a vector space $\Gamma(\mathcal{E})$ of global sections of $\mathcal{E}$. In particular,

$$\Omega^(X) = \Gamma(\Lambda^k(\mathrm{T}^* X))$$

is the space of k-forms.